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Blast Wall Analysis under Dynamic Loads: Bridging Safety and Computational Efficiency

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Abstract

Blast walls are essential for ensuring offshore facility safety during explosion events. Since few major accidents on platforms and drilling ships, blast wall designs have significantly evolved, with corrugated configurations now being widely adopted. This paper presents a computationally efficient finite element approach for evaluating the dynamic behavior of a corrugated blast wall designed for a major oil and gas installation in the Middle East. Two transient analysis methods—full transient and modal superposition (MSUP)—are employed to simulate the wall's response to blast loading. The MSUP method, leveraging modal decoupling and transfer function construction, demonstrates substantial reductions in computational time while maintaining high accuracy. This comparative study highlights the MSUP approach as a reliable and efficient alternative for dynamic structural analysis, offering significant benefits in engineering productivity without compromising safety.

Introduction

For safety-critical zones with potential hazards such as explosions, chemical spills, or fire outbreaks, the implementation of engineered barriers and rated firewalls is essential to achieve effective compartmentalization and risk mitigation. Blast walls are designed to absorb and attenuate overpressure generated by explosions within confined spaces, thereby preventing the transmission of shock waves and mitigating secondary damage to adjacent areas. Thus, an appositely engineered blast wall is crucial for any offshore facility safety during explosions.

Major accidents on drilling ships and platforms between 1979 and 2015—such as Piper Alpha, Deepwater Horizon, and SOCAR—were caused by flooding, explosions, fires, and oil spills, as documented in historical databases, DNV GL [1]. One such incident is the Piper Alpha in which the platform lacked blast-resistant walls capable of withstanding explosion overpressures. The initial gas leak induced blast compromised fire-rated barriers, which were not designed for blast loads. Inadequate segregation of hazardous equipment from critical areas like living quarters and the control room significantly contributed to the scale of the disaster. Since this incident in 1988, blast wall designs have evolved. The most prevalent types of blast walls have been explained by Zhao and Cheng [2]. The type II blast wall with corrugated skin is selected for an offshore installation designed for a large oil and gas project in Middle East.

An explosion can be characterized as a highly transient dynamic loading followed by a rapid release of energy, typically in the form of a shock wave, which interacts with a structure over a very short duration. The structural response to such an event is governed by dynamic analysis principles, particularly those related to impulsive and blast loading. This paper presents a computationally efficient finite element method to evaluate blast wall performance under dynamic loads in an event of explosion, along with deformation criterion check as per API RP 2FB [3].

This paper considers a single type of blast wall with specified blast loads. To ensure robustness, a wider range of loads acting over all the types of blast walls can be further examined in future.

Problem Statement

A blast wall protects the offshore facility from escalation of events and safe-guards the essential and safety-critical items or equipment which upon failure may lead to catastrophe. To ensure the safety of any such offshore facility, the accurate design of a blast wall is of utmost necessity. Along with accuracy, computational efficiency is also to be kept in check to analyze a blast wall exposed to transient dynamic blast loads. Reportedly, two methods for transient analysis have been studied and the response and computational efficiency have been compared in this paper.

A 16.5 m long, 4.8 m high and 6 mm thick corrugated blast wall located at 93 mm from building wall is exposed to a blast load of 0.162 barg over 120 milliseconds. The blast is considered to occur globally. Thus, the blast load is applied in the entire wall. The geometry of the blast wall has been shown in Figure 1.

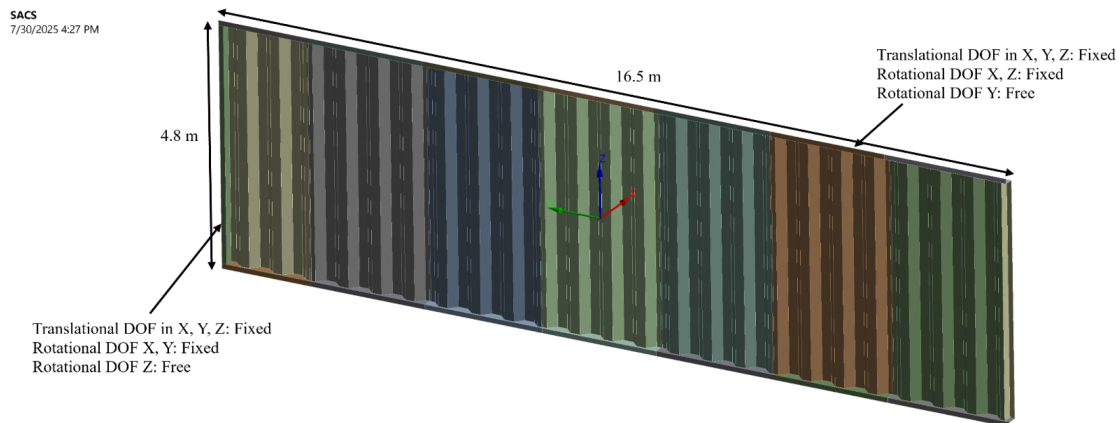


Figure 1—Blast wall model dimensions and boundary conditions considered [DOF: Degrees of Freedom]

A blast is a transient phenomenon that occurs for a very short period. The response of the structure exposed to blast load can be obtained during the blast and sometime after the blast. To obtain the response of the wall due to transient blast loads, the transient analysis is solved in two methods, namely:

- Full Transient Method
- Modal Superposition (MSUP) – Transient Method

Both the methods have been computed using commercial finite element solver ANSYS V2023 R1.

Experimental results by Bae and Paik [4] show that overpressure loads peak rapidly and decay sharply during explosions. The rise time to peak pressure varies with structural congestion—shorter rise times occur in denser configurations. After the peak, pressures drop below ambient before stabilizing. The pressure profile can be categorized into arrival time, positive phase, and negative phase.

The analysis is computed for 400 steps, obtaining result at every 1 millisecond. Accordingly, for the blast that occurred up to 120 milliseconds, response of the wall is observed until 400 milliseconds.

Structural Model

The blast wall structural analysis is performed for a 16.5 m long, 4.8 m high and 6 mm thick corrugated blast wall located at 93 mm from building wall. The 3D model of the blast wall used for this analysis is presented in [Figure 1](#).

FEA model with time-history analysis using ANSYS software is used to carry out the local blast wall checks. The local model consists of the crimp plate and angle between crimp plate to the adjacent structure. Quadrilateral shell elements are used in the FEA model. A uniform quadrilateral mesh with element size of 50 mm and 30 mm is chosen for the corrugated blast wall and angle section FEA model respectively.

Boundary Conditions

The boundary conditions at top and bottom of the angle section are restrained in all the three (3) translation directions and rotation in X and Z directions, as represented in [Figure 1](#). Two (2) sides of the wall model are restrained in all the three (3) translation directions and rotation in X and Y directions, as represented in the same figure. The boundary conditions replicate the actual restrained condition of the wall.

Methodology

Structural analysis under dynamic loading can use either a full transient methodology or a modal superposition (MSUP) approach. The former involves solving the complete set of dynamic equations non-linearly, while the latter uses linear superposition of modes.

The MSUP approach involves decoupling of the system equations and solving n number of decoupled single-degree-of-freedom systems instead of a single coupled n-order multiple degrees of freedom system in the modal domain. Modal parameters are used to construct the transfer function, which gets coupled with the external force producing displacement field and subsequently translated to the time domain.

Full Transient Approach

Transient dynamic analysis, commonly referred to as time-history analysis, is a computational technique employed to evaluate the dynamic response of a structure subjected to arbitrary time-dependent loads. This method facilitates the determination of temporal variations in structural parameters such as displacements, strains, stresses, and internal forces.

The formulation of transient dynamic analysis integrates the mass and damping properties of the structure into the governing equations of motion, enabling the simulation of its response to combinations of static, transient, and harmonic excitations. This approach ensures that the time scale of loading and the corresponding dynamic characteristics are appropriately accounted for.

In cases where inertial and damping effects are negligible, simplified static analysis methods may suffice; however, for high-fidelity modeling of dynamic phenomena, transient dynamic analysis remains indispensable.

The basic equation of motion solved by a transient dynamic analysis is:

$$\{F(t)\} = [M]\{\ddot{u}\} + [C]\{\dot{u}\} + [K]\{u\} \quad \text{Eq. 1}$$

where:

$\{F(t)\}$ = Load vector	$\{\ddot{u}\}$ = Nodal acceleration vector
$[M]$ = Mass matrix	$\{\dot{u}\}$ = Nodal velocity vector
$[C]$ = Damping matrix	$\{u\}$ = Nodal displacement vector
$[K]$ = Stiffness matrix	

At any given time, t , these equations can be thought of as a set of "static" equilibrium equations that also consider inertia forces ($[M]\{\ddot{u}\}$) and damping forces ($[C]\{\dot{u}\}$). The ANSYS program allows the Newmark time-integration method, or an improved method called HHT to solve these equations at discrete time points. The time increment between successive time points is called the *integration time step*.

Unlike static analysis, which assumes instantaneous equilibrium and neglects inertial and damping effects, transient dynamic analysis captures the influence of inertia and energy dissipation mechanisms, making it essential for scenarios where the rate and duration of loading significantly affect structural behavior.

Modal Superposition Approach (MSUP)

This method is efficient for linear systems; reduces computational cost by truncating higher modes. The equation of motion is transformed into modal domain and works based on linear superposition of different modes. This is quite efficient in terms of computational cost. It requires less memory also. However, in general understanding, as it is a linear method, it restricts inclusion of different non-linearities.

The procedure begins with a free vibration analysis – referred to as modal analysis—to determine time-independent modal parameters such as natural frequencies and mode shapes. These parameters are used to construct the frequency response function, which further gets coupled with the external force to produce displacement field in the time domain. In ANSYS, performing modal analysis is a prerequisite for conducting modal superposition (MSUP) transient analysis. To improve the accuracy of MSUP results, it is essential to consider a higher number of modes, along with a refined mesh and smaller time step. In the present study, the first 80 fundamental modes have been included to meet these requirements. A comprehensive comparison is also performed for the given blast-wall configuration and the results are reported.

Table 1—Differences in two methodologies

Full Transient Analysis	Modal Superposition Method
For blast wall analysis, full transient analysis simulates the dynamic response of the wall to an explosion or blast wave over time.	In blast wall analysis, the modal superposition method utilizes the wall's eigenmodes to analyze its response to a blast load.
It captures the wall's behavior throughout the entire blast event, accounting for all time-dependent effects.	By considering the dominant modes of vibration, it simplifies the dynamic analysis of the wall under blast loading conditions.
<i>Workflow</i>	
The blast wall geometry, material properties, support conditions, and blast load characteristics are defined. The transient analysis is set up, specifying the time range, time step, and blast wave parameters. The analysis is solved to obtain detailed time histories of the wall's dynamic response to the blast.	A modal analysis of the blast wall is performed to determine its natural frequencies, mode shapes, and modal masses. The blast loading scenario is defined, including the blast impulse or pressure-time history. The modal superposition method is applied to calculate the wall's response by combining mode shapes and modal participation factors.
<i>Solving Methodology</i>	
The dynamic equations of motion are solved for the blast wall subjected to the blast load, considering inertial, damping, and stiffness effects over time.	The response of the blast wall is expressed in terms of a linear combination of mode shapes weighted by modal participation factors. The dominant modes influencing the wall's dynamic behavior under blast loading conditions are determined.
<i>Accuracy and Computational Time</i>	
Deflection Value: 16.255 mm	Deflection Value: 15.473 mm
Computational Time: High	Computational Time: Low

Full Transient Analysis	Modal Superposition Method
<i>Fundamental difference – Summary</i>	
The full transient analysis obtains the displacement values in nodal domain. Any structure is discretized into finite number of elements. In transient analysis, the response due to force at every node is calculated. So, if any structure is having, for example, 1 million nodes and the number of time steps is 1000 then it will generate a massive system which is computationally intensive to solve.	In MSUP method, the dynamic equilibrium equation is decoupled for each mode. It is solved separately for each mode and then linear superposition method is applied to obtain the combined modal displacement field and finally it is transferred to the nodal displacement. As superposition of modes fetch the displacement of the structure, so more the number of modes, more accurate the result. In this study, the first 80 modes are used which fetched quite accurate results.

Load considerations

Any structure is exposed to main three types of loading, namely harmonic loading, non-harmonic deterministic loading and non-harmonic non-deterministic loading or random excitation.

- *Harmonic excitation* occurs at a particular frequency and the loading has a definite pattern of response, that is, response is steady.
- *Non-harmonic deterministic loading* cannot be represented with a sine or cosine function unlike harmonic loading but can be defined as a time-dependent function.
- *Non-harmonic non-deterministic loading* or *random excitation* cannot be defined as a time function, that is it varies randomly with respect to time.

Due to varying explosion conditions, the load profile is complex. Accurately characterizing blast pressure requires tracking its pressure-time history, including rise time, peak pressure, duration, and decay behavior, as described by Jung et. al. [5].

The blast design loads are generated in near accordance with the explosion risk analysis report of the corresponding project. The explosion loads considered for the analysis are provided in Table 2. The blast load is then applied in time history blast input based on blast load curve/profile. Besides the blast loads, gravity loads have also been considered.

Table 2—Explosion Loads on Blast Wall

Component	Blast Overpressure (barg) ^{1)*}
Fire/ Blast wall – Loads	0.162 (120ms)

Note:

1. Unit barg: the ‘g’ represents gauge pressure which is the pressure relative to atmospheric pressure. For the pressures above atmospheric pressure, gauge pressure is positive.
2. Number in bracket () is blast impulse duration.

The blast duration is based on the triangular load profile as shown in Figure 2. The time to reach peak overpressure shall be 0.5 times the total duration of blast.

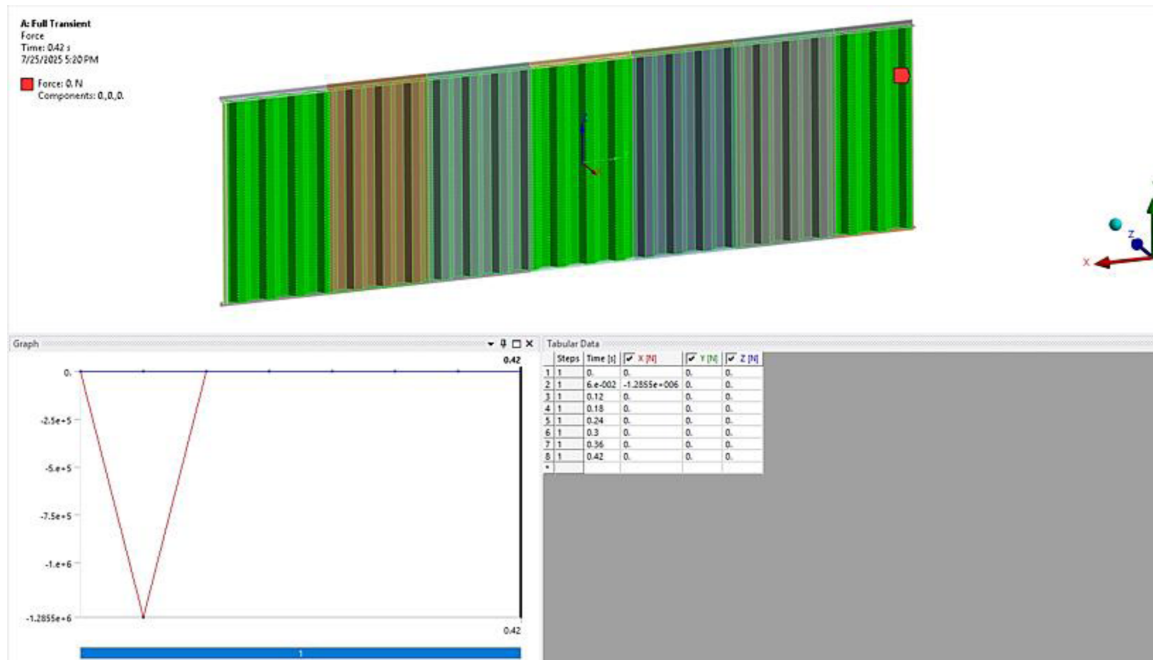


Figure 2—Global Blast Load Profile

The global blast load profile is shown in Figure 2. The negative sign refers to the direction of the application of load as per the coordinate system chosen. The global blast pressure is considered acting on overall area of the blast wall.

Results

A 16.5 m long, 4.8 m high and 6 mm thick corrugated blast wall at 93 mm from building wall, exposed to a blast load of 0.162 barg over 120 milliseconds is computationally analyzed using the full transient and MSUP-transient methods. Both the methods have been computed using commercial finite element solver ANSYS V2023 R1.

Transient analysis results

Transient dynamic analysis was performed over a total duration of 400 milliseconds, with the global blast load applied for the initial 120 milliseconds. A time step of 1 millisecond was used, resulting in 400 discrete time points for evaluating the structural response, as illustrated in the displacement plot for global blast pressure (Figure 4). The node experiencing the maximum magnitude of displacement in the direction of the blast load was identified (Figure 3), and its time-dependent displacement response was subsequently extracted and plotted (Figure 4). The displacement curve during and immediately following the blast event exhibits characteristics indicative of damping behavior.

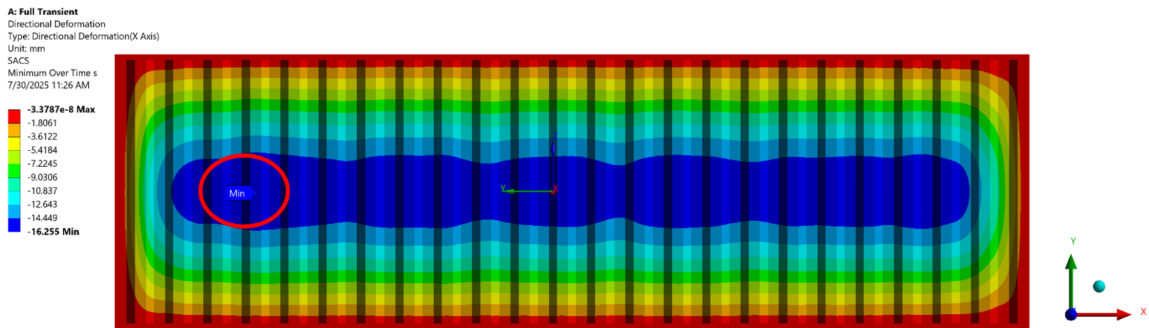


Figure 3—Maximum magnitude of deformation over time along blast load direction: Transient Analysis

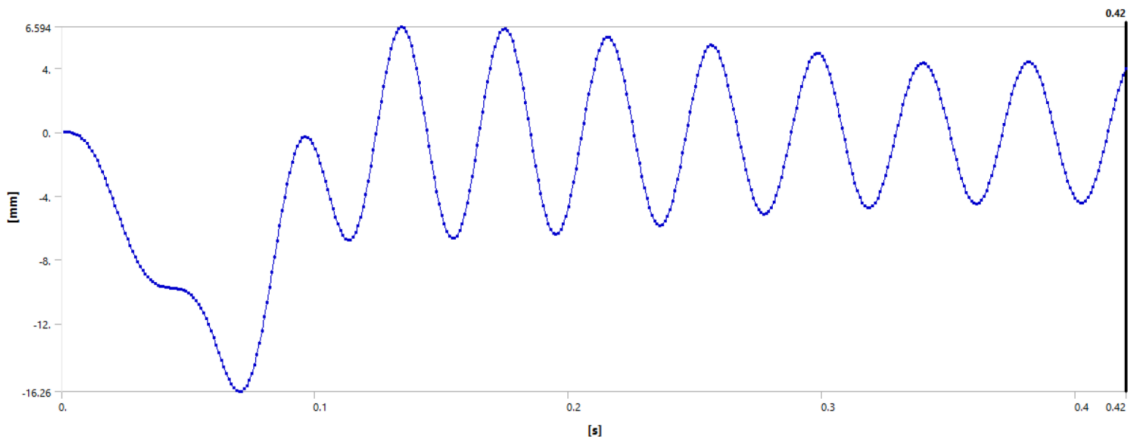


Figure 4—Displacement Plot over time for Global Blast Load at the node where maximum magnitude of deformation is observed: Transient Analysis

The complete transient analysis was executed on a system equipped with an 11th Gen Intel® Core™ i7-11850H processor (2.50 GHz) and 32 GB RAM. The total computational time required for the simulation was approximately 1 hour and 41 minutes, as illustrated in Figure 5.

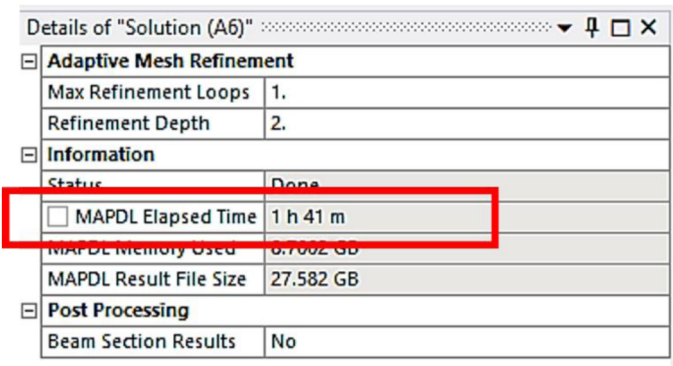


Figure 5—Time required for completion of full transient analysis

MSUP Approach results

Step-1: Modal Analysis. The first 80 natural modes were computed and superposed within the linear transient analysis framework to determine the wall's maximum deflection over time. The corresponding modal frequencies are presented in Figure 6. Although a theoretically exact solution using the Modal Superposition method would require an infinite number of modes, such an approach is computationally

impractical. Therefore, a set of 80 modes was selected, providing a close approximation to the ideal response.

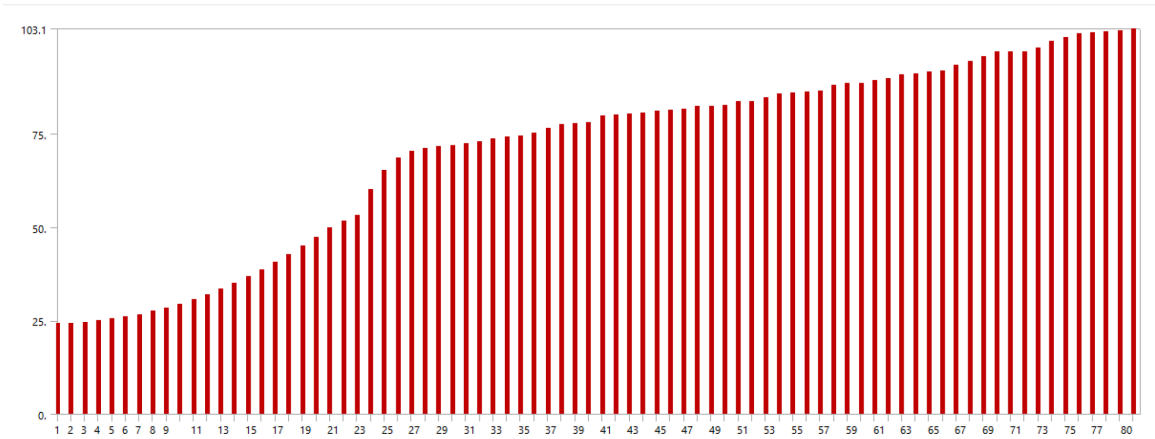


Figure 6—Frequency of first 80 modes, obtained from modal analysis

The modal analysis was performed on the same system configuration used for the transient analysis—an 11th Gen Intel® Core™ i7-11850H processor (2.50 GHz) with 32 GB RAM. The total computation time for extracting the first 80 modes was approximately 1 minute and 16 seconds, as shown in Figure 7.

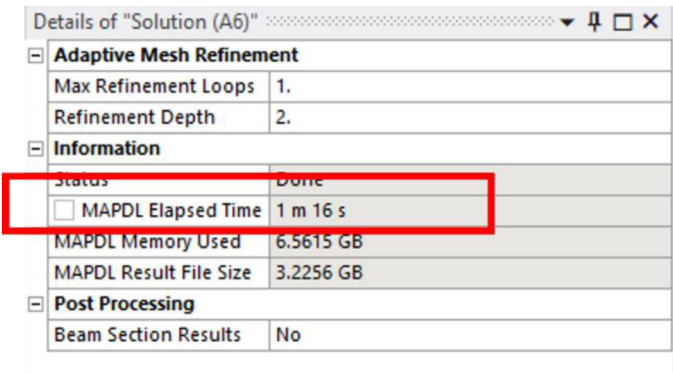


Figure 7—Time required for completion of Modal analysis

Step-2: MSUP Transient Analysis. In the modal superposition (MSUP) approach, the first 80 modes were computed and utilized in the transient analysis to evaluate the structural response. The simulation was carried out over a duration of 400 milliseconds, with the global blast load applied for the initial 120 milliseconds. The node exhibiting the maximum displacement in the direction of the blast load was identified (Figure 8), and its time-dependent response was analyzed (Figure 9).

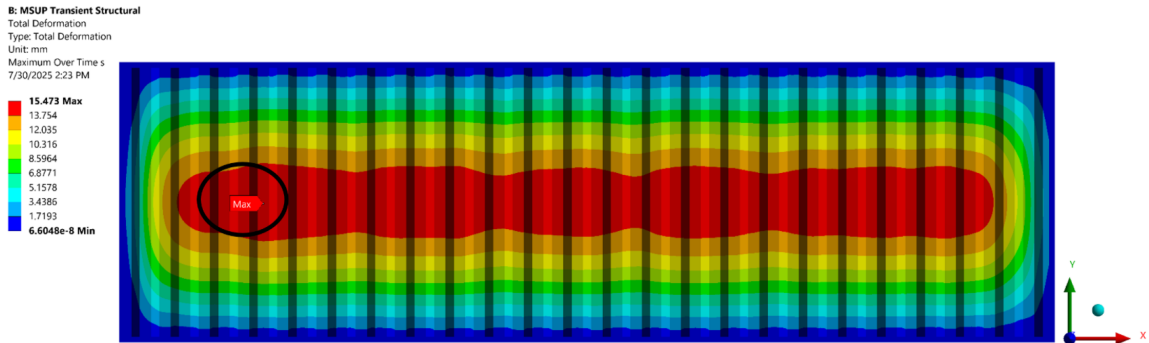


Figure 8—Maximum magnitude of deformation over time along blast load direction: MSUP Transient Analysis

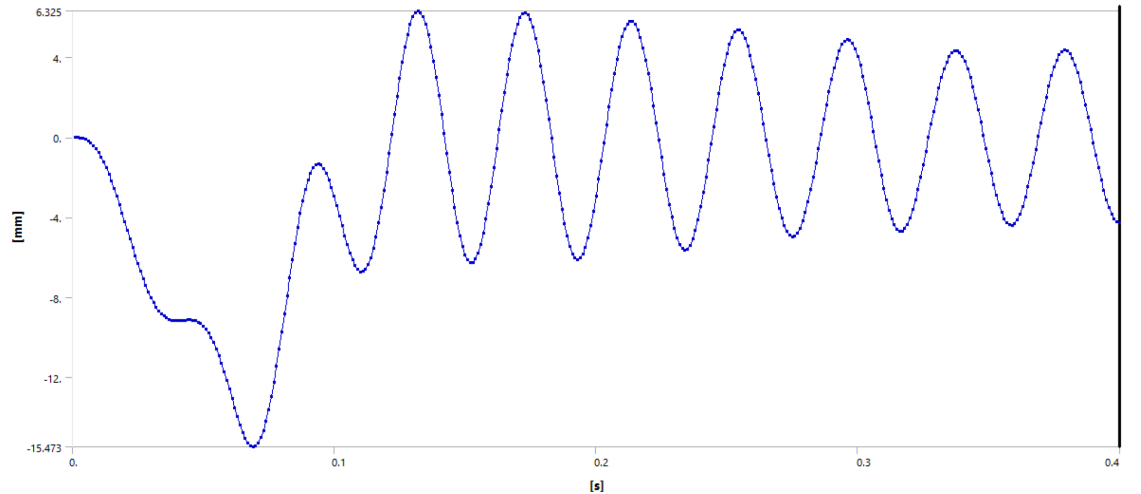


Figure 9—Displacement Plot over time for Global Blast Load at the node where maximum magnitude of deformation is observed: MSUP Transient Analysis

The total computation time required to complete the MSUP approach transient analysis is 5 minute and 8 seconds, in the same configured device as mentioned for transient analysis, is shown in Figure 10.

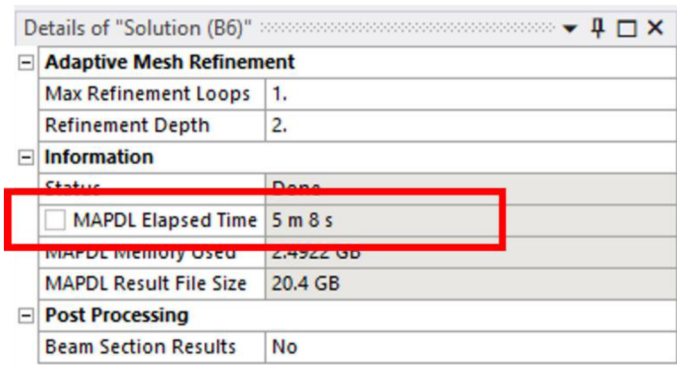


Figure 10—Time required for completion of MSUP approach transient analysis

Results from the blast wall analysis confirm that all structural components satisfy the design criteria under global blast pressure conditions. A comparative summary of the results obtained from both analysis methods is presented in Table 3.

Table 3—Results Summary

Analysis Methodology	Maximum Deflection (mm)	Back bounce Deflection (mm)	Computational Time
Full Transient Analysis	16.255	6.594	1 hour 41 minutes
MSUP Transient Analysis	15.473	6.325	6 minutes 24 seconds
Error Quantification (%)	4.811	4.079	93.67

Based on the outcomes of the simulations, the blast wall demonstrates sufficient capacity to withstand the pressure generated during the blast event and not deform up to the building wall, at 93 mm from blast wall. Thus, the deformation of the blast wall is limited, stating that ‘the wall remains in-place without rupture or disconnection from their supports’, as per API RP 2FB [3].

The full transient method yielded a maximum deflection value of 16.255 mm in the considered time, with a computational time of 1 hour 41 minutes. The MSUP method yielded deflection value as 15.473 mm, using 80 modes, with a total computational time of 6 minutes and 24 seconds. Considering the full transient method as the true value, the error quantification between the two methods is also shown in Table 3.

Hence, it can be observed that, there is a reduction in computational time by almost 94% and the computationally efficient MSUP approach yields results in great agreement with the conventional full transient method, with less than 5% difference.

Conclusions

For safety-critical zones prone to hazards like explosions, chemical spills, or fires, engineered barriers and rated firewalls are vital for effective compartmentalization and risk mitigation. Blast walls absorb and attenuate explosion overpressure, preventing shock wave transmission and protecting adjacent areas. Properly designed blast walls are essential for offshore facility safety during such events.

This study contributes to that evolving body of knowledge by presenting a computationally efficient finite element approach for evaluating the dynamic response of blast walls under transient explosive loads. By comparing the Full Transient Method and the Modal Superposition (MSUP) Method, the research demonstrates that the latter offers a remarkable reduction in computational time—up to 94%—without compromising the accuracy or safety of the structural response. This efficiency is particularly valuable in large-scale engineering projects where time and resource optimization are crucial.

The selected corrugated blast wall exemplifies the practical application of advanced analysis techniques in real-world scenarios. The study's findings not only validate the structural integrity of the wall under specified blast conditions but also establish a reliable framework for future design and assessment practices.

In conclusion, the integration of computational efficiency with structural accuracy marks a significant step forward in offshore safety engineering. The methodology presented here can serve as a benchmark for future blast wall designs, ensuring enhanced protection of critical infrastructure and personnel in hazardous environments.

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